

SynapseAI

Your personal brain-age report - clear data, model context, and published research

YOUR AGE	MODEL ESTIMATE	DIFFERENCE
54 years	47.3 years	6.7 years lower

Estimated brain age is about 6.7 years younger than chronological age.

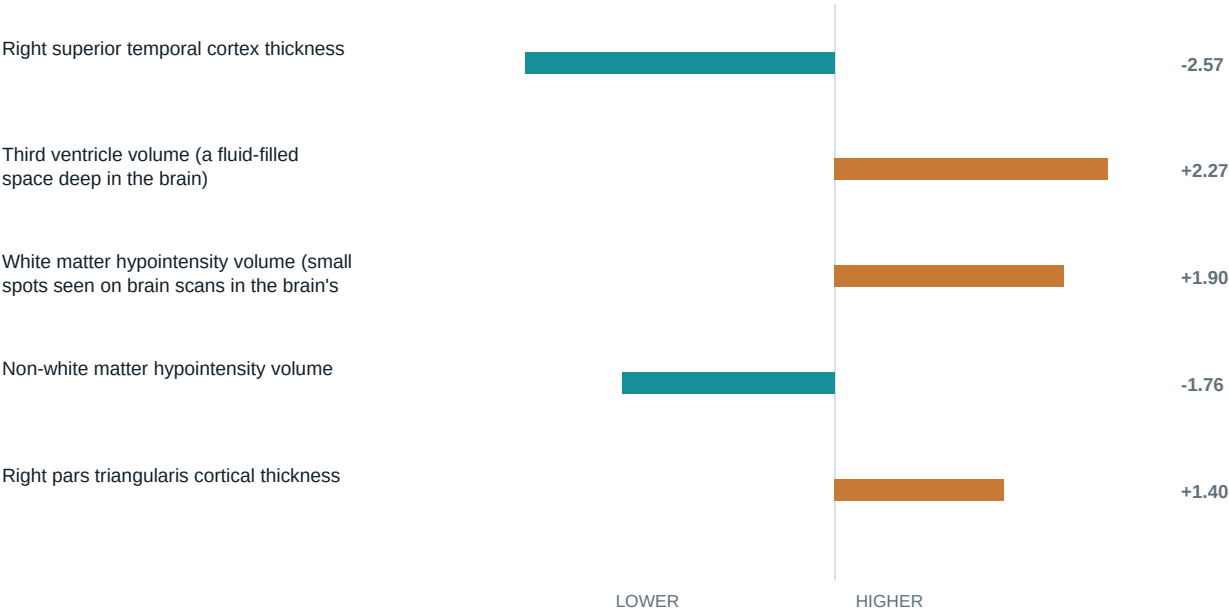
This person's chronological age is 54, but the model's estimate based on brain scan patterns was about 47. Some individual measurements, like the thickness of a right-side hearing-related brain area and a certain type of scan spot, leaned toward a younger-appearing estimate, while others, such as the size of a small fluid-filled space and some scan spots in the brain's wiring, leaned toward an older-appearing estimate. The overall younger-leaning result reflects a combination of these different, sometimes opposing, signals rather than any single measurement acting alone.

KEY TAKEAWAY

The overall younger-leaning estimate reflects a personal pattern of measurements pulling in different directions, with the clearest insight coming from viewing them together.

What influenced the estimate

Bars show how strongly each MRI-derived measurement moved this model estimate lower or higher.



01 Right superior temporal cortex thickness

Younger-leaning	Reference z: +0.68	Evidence: Moderate
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WHAT THIS MEASUREMENT IS

The superior temporal cortex is a strip of brain tissue on the side of the brain, near the ear, involved in processing sounds and language. 'Cortical thickness' simply means how thick this layer of brain tissue is at that location.

WHAT THE MODEL SAW

This measurement pushed the model's prediction toward a younger brain age. The model treated a thicker right superior temporal cortex as a sign associated with lower predicted brain age in this analysis.

WHAT IT MEANS IN THIS RESULT

This measurement was somewhat above the reference average, and the model treated it as one of several factors linked with a younger predicted brain age. Research shows that thickness in this brain region varies naturally between people for reasons such as age and biological differences. Here, it offers one piece of context within the person's broader result.

WHAT PUBLISHED RESEARCH ADDS

Research in healthy older adults shows that cortical thickness, including in temporal regions, tends to decrease with normal aging, though the pattern varies by brain region. One study found that cognitively healthy older adults of different ethnic backgrounds showed natural variation in thickness in this exact region, suggesting normal, non-clinical differences exist. Another study in healthy elderly people found that a specific genetic variation was linked to a thicker superior temporal cortex, showing that thickness in this area can vary for reasons unrelated to disease. Together, these studies suggest thickness in this region can differ across healthy individuals for several reasons, including age and normal biological variation.

The supplied studies are relevant to healthy aging and this exact brain region, but they are based on different research questions and populations, so the connection to this individual's result is indirect.

AMASS RESEARCH SOURCES

- Cortical thickness is associated with different apolipoprotein E genotypes in healthy elderly adults - doi:10.1016/j.neulet.2010.05.092
- Differences in cortical structure between cognitively normal East Asian and Caucasian older adults: a surface-based morphometry study - doi:10.1038/s41598-020-77848-8
- Normal age-related brain morphometric changes: nonuniformity across cortical thickness, surface area and gray matter volume? - doi:10.1016/j.neurobiolaging.2010.07.013

02 Third ventricle volume (a fluid-filled space deep in the brain)

Older-leaning	Reference z: +1.42	Evidence: Moderate
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WHAT THIS MEASUREMENT IS

The third ventricle is a small fluid-filled space located near the center of the brain. Its size can change somewhat as people age.

WHAT THE MODEL SAW

This measurement pushed the model toward predicting an older brain age. It was one of several factors the model weighed, and its influence should not be read as a specific number of years.

WHAT IT MEANS IN THIS RESULT

The model detected that this person's third ventricle volume is somewhat larger than average for the reference group used (about 1.4 standard deviations above the mean), and this was linked with an older-leaning prediction. Research on general aging suggests that this kind of enlargement can be part of typical brain changes over time. It is most useful as one part of the broader measurement pattern.

WHAT PUBLISHED RESEARCH ADDS

General aging research included in this review found that brain ventricles, including this one, tend to become larger with increasing age, though the pattern is not always simple or straight-line. This suggests some enlargement of these spaces is a common part of aging in general, not something unusual on its own. Other studies in the retrieved evidence focused on specific clinical populations and are not detailed here because they do not directly describe healthy aging.

One study directly supports a link between ventricle size and general healthy aging, but it discusses ventricles broadly rather than the third ventricle specifically, and does not establish a precise threshold for this individual's value.

AMASS RESEARCH SOURCES

- Consistent neuroanatomical age-related volume differences across multiple samples -
doi:10.1016/j.neurobiolaging.2009.05.013

03 White matter hypointensity volume (small spots seen on brain scans in the brain's internal wiring)

Older-leaning	Reference z: -0.17	Evidence: Moderate
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WHAT THIS MEASUREMENT IS

White matter is the brain's internal wiring that connects different regions together. 'Hypointensities' are small spots seen on certain MRI scans that show up in this wiring, and they are commonly looked at in brain-aging research.

WHAT THE MODEL SAW

This factor pushed the model's prediction toward an older brain age. However, the patient's own measurement was close to the reference average, so this factor's influence does not reflect an unusual or extreme value for this patient.

WHAT IT MEANS IN THIS RESULT

The model treated this measurement as contributing to an older-appearing brain-age estimate, even though the person's value was near the average seen in the reference group. Research shows these spots appear commonly with age and relate only loosely to overall brain health, so this measurement is most useful as part of the combined pattern.

WHAT PUBLISHED RESEARCH ADDS

Research in adults without dementia has found that these small white-matter spots become more common with age and are linked to subtle changes in the microscopic structure of nearby brain wiring. Studies also report that white matter volume in general tends to decline gradually with age. However, the research notes that these spots only loosely relate to a person's overall clinical status, and age-related changes in brain wiring can occur even without them.

The supplied studies focus on healthy or nondemented aging populations and consistently discuss this feature in relation to general aging, but they describe loose, not strong, associations.

AMASS RESEARCH SOURCES

- Associations between T1 white matter lesion volume and regional white matter microstructure in aging - doi:10.1002/hbm.22236
- Sex Differences in Cortical Morphometry and White Matter Microstructure During Brain Aging and Their Relationships to Cognition - doi:10.1093/cercor/bhab155
- White matter in aging and cognition: a cross-sectional study of microstructure in adults aged eighteen to eighty-three - doi:10.1080/87565641003696775

04 Non-white matter hypointensity volume

Younger-leaning	Reference z: -0.43	Evidence: Insufficient
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WHAT THIS MEASUREMENT IS

This measure looks at small darker spots seen on brain scans that are not part of the brain's white matter (the wiring that connects brain regions). These spots can show up in different brain areas and are sometimes studied as markers of brain changes over time.

WHAT THE MODEL SAW

This measurement pushed the model's prediction toward a younger brain age. The size of that push does not translate into a specific number of years.

WHAT IT MEANS IN THIS RESULT

This patient's non-white matter hypointensity volume was close to the reference average (z-score of about -0.43), and it nudged the model's prediction toward a younger estimated brain age. Because the available research does not closely match this exact measurement, we cannot say much more about what this specific result means for this person's brain aging.

WHAT PUBLISHED RESEARCH ADDS

The scientific papers supplied mostly discuss general white matter and gray matter changes with aging, not this specific hypointensity measurement. One paper looked at hypointensities in a specific brain structure (the caudate nucleus) in older non-demented adults and linked more widespread hypointensity there to other age-related brain changes, but this is a different, narrower measurement than the one used in this model. Because of this mismatch, the research does not directly explain what this patient's specific value means.

The supplied studies focus on white/gray matter volume changes or on a specific different brain structure (caudate nucleus), not on this general non-white matter hypointensity measurement, so they do not directly support an interpretation of this feature.

AMASS RESEARCH SOURCES

- Caudate nucleus hypointensity in the elderly is associated with markers of neurodegeneration on MRI -
doi:10.1016/j.neurobiolaging.2007.05.008

05 Right pars triangularis cortical thickness

Older-leaning	Reference z: -0.08	Evidence: Insufficient
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WHAT THIS MEASUREMENT IS

The pars triangularis is a small part of the frontal lobe, the front part of the brain. Cortical thickness measures how thick the outer layer of brain tissue is in that area.

WHAT THE MODEL SAW

This brain measurement pushed the model's prediction toward an older estimated brain age. The measurement itself was close to the reference average, so this influence was modest rather than extreme.

WHAT IT MEANS IN THIS RESULT

The model found that this measurement had a small effect pushing the predicted brain age slightly older, but the patient's own value was close to the reference average for this dataset. Since the available research does not include healthy-aging studies of this specific region, we cannot say what this pattern means for this person's brain aging in a general sense.

WHAT PUBLISHED RESEARCH ADDS

The evidence available for this specific brain region comes only from studies of particular clinical populations, not from studies of healthy aging. Because of this, the supplied research cannot be used to explain what this measurement means for typical, healthy brain aging.

All supplied studies focus on specific clinical populations rather than healthy aging, so they do not provide adequate evidence to interpret this feature in a general population.

How to use this report

SynapseAI brings your brain-age estimate, influential measurements, and research context into one personalized explanation. It is designed to support understanding and informed conversations with qualified professionals.

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